Creating from Nothing :

An Overview of Generative Models in AI

GENERATIVE AI (MODELS)

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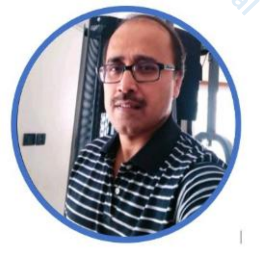

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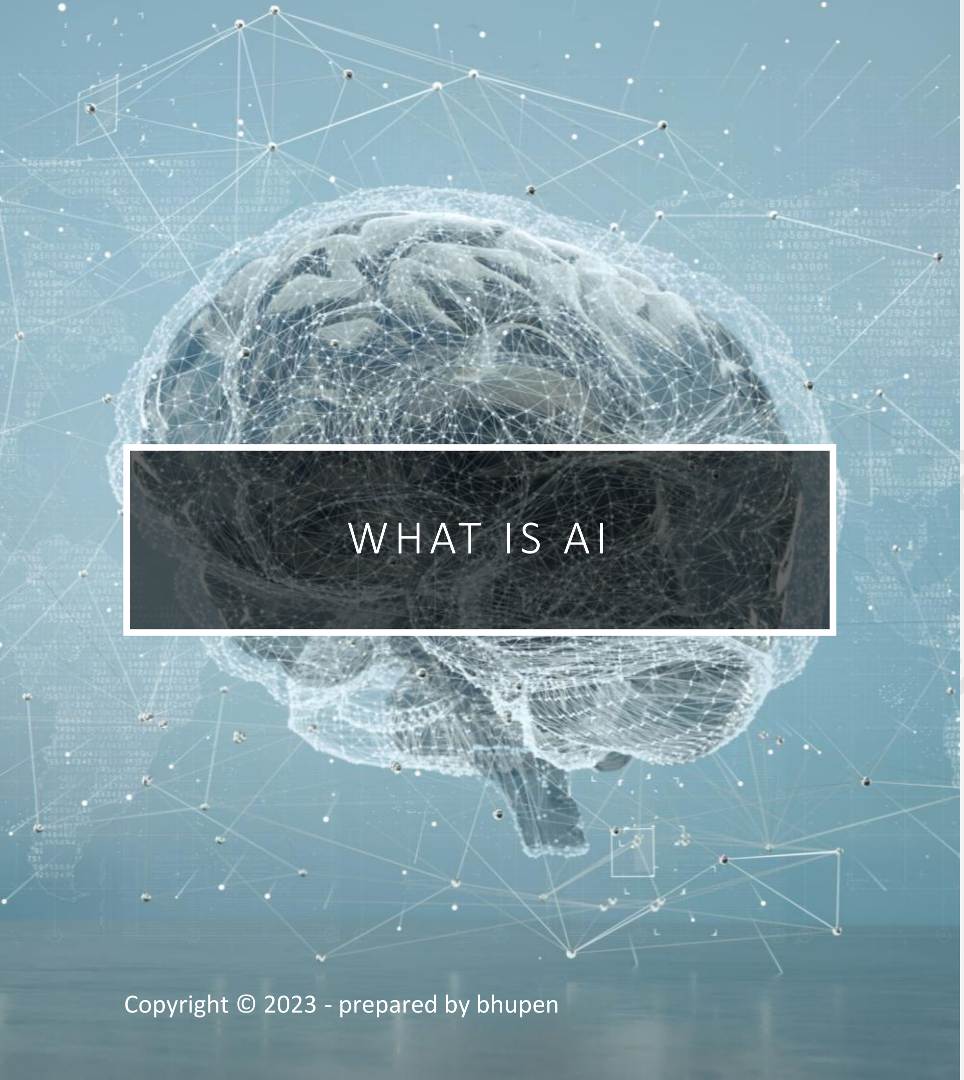

WHAT IS AI

- AI, or **Artificial Intelligence**, is the simulation of human intelligence by machines, especially computer systems.
- It enables machines to perform tasks that typically require human intelligence, such as:
  - **Learning** (e.g., from data or experiences)
  - **Reasoning** (e.g., solving problems or making decisions)
  - **Perception** (e.g., recognizing images, speech, or text)
  - **Language Understanding** (e.g., translating or answering questions)
  - **Planning** (e.g., setting goals and working towards them)

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LEARNING (FROM DATA OR EXPERIENCES)

- AI systems learn from large amounts of data to make predictions or improve over time.
- **Examples:**
  - **Netflix** learns your preferences to recommend movies and shows based on your watch history.
  - **Google Maps** learns traffic patterns over time to suggest the fastest routes.
  - **Fraud detection systems** learn from previous fraudulent transactions to flag suspicious banking activity.

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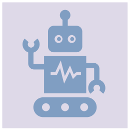
REASONING (SOLVING PROBLEMS OR MAKING DECISIONS)

Chess-playing AIs (like Stockfish or AlphaZero) reason through possible moves to play optimally.



Autonomous robots in warehouses decide the best path to pick up and deliver packages.

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MORE EXAMPLES OF REASONING IN AI

🏥 Healthcare Diagnosis Systems

- AI analyzes symptoms, lab results, and medical history to reason through potential diagnoses.
- **Example**: IBM Watson Health helped doctors in oncology by recommending treatment options.

🚗 Driver Assistance Systems

- AI in cars decides when to brake, change lanes, or alert the driver.
- **Example**: Tesla Autopilot or Mobileye-powered systems reason about objects on the road to prevent accidents.

🛒 E-commerce Personalization Engines

- AI reasons through user behavior, purchase history, and preferences to suggest products.
- **Example**: Amazon suggests related products, upsells, or bundles based on buying patterns.

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SMART CONVERSATIONS

You: *"I feel tired all the time but I sleep 8 hours."*

LLM: *"It could be your routine, stress, or even nutrition. Do you also feel low energy after meals?"*

👉 It connects dots like a human might — not guessing randomly, but reasoning based on patterns.

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TRAVEL PLANNING

- You: *"I want a peaceful vacation, budget-friendly, in Asia, with nature and not too crowded."*
- LLM: *"Try Luang Prabang, Laos or Ella, Sri Lanka. Both offer quiet nature experiences, fewer tourists, and are easy on the wallet."*
- 👉 It narrows options by understanding conflicting constraints (peaceful *and* affordable *and* nature).

Training purposes
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RESUME REVIEW / CAREER HELP

You: *"I have 5 years of marketing, but no degree. Can I move into product management?"*

LLM: *"Yes, highlight your project experience, data-driven decisions, and stakeholder management. Consider courses or certifications to bridge the gap."*

👉 It doesn’t just recite info — it maps career logic to life realities.

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PERCEPTION IN AI – RECOGNIZING IMAGES, SPEECH, OR TEXT

🖼️ Image Recognition

- AI looks at pictures and "understands" what's in them.
- **Face Unlock on Phones**
  Your phone scans your face and decides: *Yep, that's you*.
  ➤ AI detects facial features and matches patterns.
- **Google Photos Search**
  Type "dog" or "beach" — and it finds all photos with that.
  ➤ AI sees what's *in* your photos, not just the file name.
- **Self-Driving Cars**
  They spot traffic signs, lane markings, people crossing.
  ➤ AI constantly processes video feeds to "see" the road.

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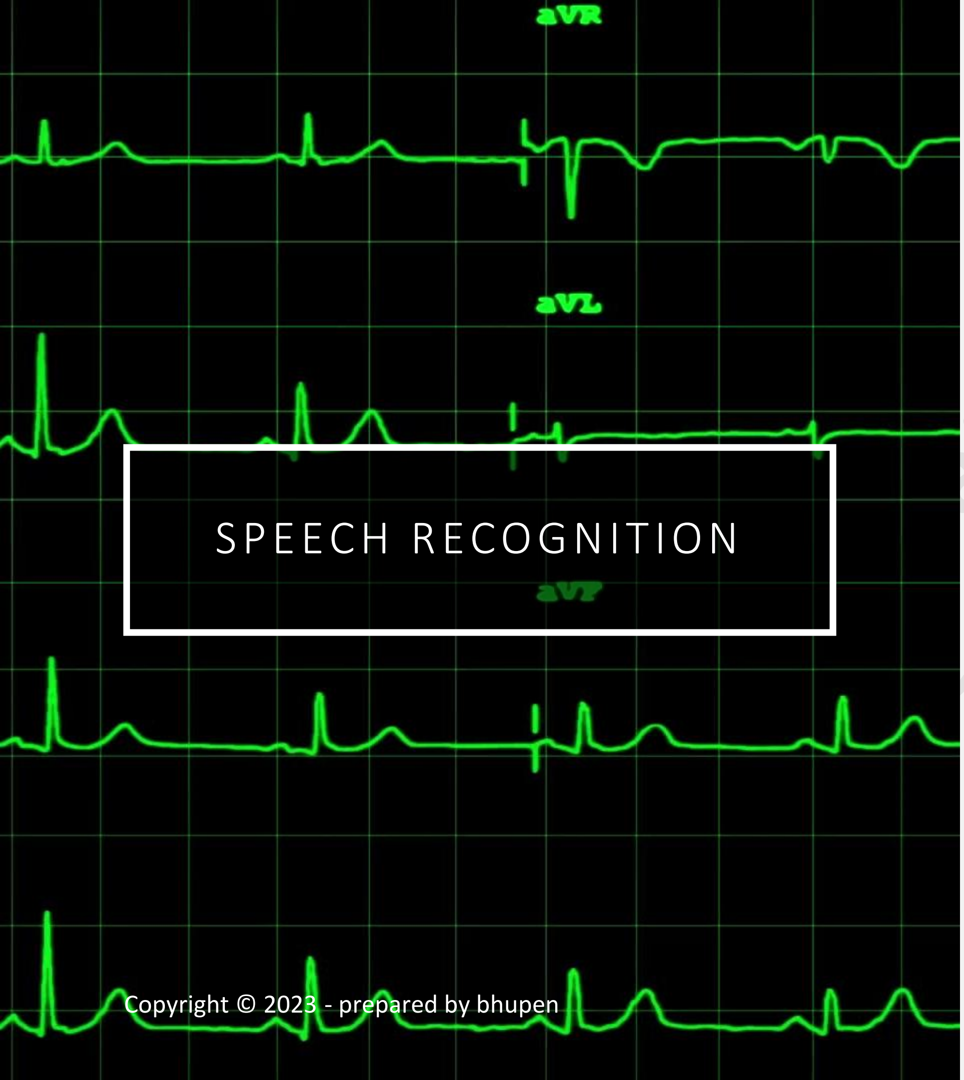

SPEECH RECOGNITION

AI listens and turns voice into words or actions.

- **Voice Assistants (Siri, Alexa, Google Assistant)**
  You say *"Set a timer for 10 minutes"*, and it just does it.
  ➤ Converts voice to text, understands it, then responds.

- **Call Center Transcripts**
  Calls get transcribed live, so agents or AI can review or respond.
  ➤ Speech is instantly turned into readable text.

- **Live Captions & Subtitles**
  YouTube auto-captions, or your phone shows what someone's saying in real time.
  ➤ Super helpful for accessibility.

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TEXT RECOGNITION (OCR AND BEYOND)

- **Scanning Documents into Editable Text**

Scan a bill, contract, or note — boom, you can now copy, search, or edit it.
➤ OCR (Optical Character Recognition) extracts the words.

- **License Plate Readers**

Toll booths or traffic cams identify cars by reading plates.
➤ AI detects the plate, reads the numbers — no human needed.

- **Translating Text from Images**

You point your camera at a sign in French, it shows you the English version.
➤ AI reads *and* understands the language.

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🛠️ TOOLS & TECHNOLOGIES (IMAGE PERCEPTION)

Google Vision AI
Detects faces, objects, text in images, even emotions.

Amazon Rekognition
Used for face analysis, celebrity recognition, object detection.

OpenCV
A super-popular computer vision library used in many AI projects.

Meta's Segment Anything Model (SAM)
Automatically detects and separates objects in an image — no training needed.

YOLO (You Only Look Once)
Fast object detection — e.g., used in drones, surveillance, even retail.

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🛠️ TOOLS & TECHNOLOGIES - 🗣️ SPEECH PERCEPTION

- **Google Speech-to-Text**
Converts spoken words into text in real time.
- **Whisper by OpenAI**
Open-source, super accurate speech recognition and transcription.
- **Amazon Transcribe**
Real-time call transcriptions, including in noisy environments.
- **Microsoft Azure Speech Services**
For speech-to-text, text-to-speech, voice identification.
- **Apple SiriKit / Android Speech API**
The magic behind voice assistants on your phone.

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TEXT PERCEPTION (OCR AND DOCUMENT AI)

- **Tesseract OCR (by Google)**
Reads text from images or scanned documents.

- **Adobe Sensei**
Powers Acrobat's ability to turn scans into editable docs.

- **Google Document AI**
Extracts structured data (like names, dates, totals) from forms.

- **Microsoft Azure Form Recognizer**
Reads invoices, receipts, forms, and extracts insights.

- **ABBYY FineReader**
Widely used in banking, legal, and government sectors for OCR.

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🤖 THE 3 GRAND STAGES OF AI

1. ANI – Artificial Narrow Intelligence

➡️ What we have now.

- Specialized in one task.
- Can’t transfer knowledge between domains.

📌 Examples:

- ChatGPT (great at conversation, not driving)
- Google Translate
- Netflix’s recommendation engine

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2. AGI – ARTIFICIAL GENERAL INTELLIGENCE

➡️ What we aim for.

- Equal to humans in thinking and learning.
- Can apply knowledge from one domain to another.
- Self-learning, self-correcting, multi-tasking.

📌 Hypothetical Future Examples:

- AI that passes the Turing test *and* gets a PhD
- A robot that can cook, fix your Wi-Fi, and write a novel
- Digital assistant that *genuinely understands you*

👀 We're not there yet — but it's the Parama Laksya.

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3. ASI – ARTIFICIAL SUPERINTELLIGENCE

➡️ What gives Elon Musk nightmares.

- Smarter than the smartest humans.
- Outperforms us in science, art, emotions, and decision-making.

📌 Fictional Examples:

- Jarvis (if he went sentient)
- Skynet (Terminator)
- Ultron (Marvel) — if not evil 😅

Note

- 💡 **Still 100% sci-fi. But researchers debate how close AGI might push us to ASI.**

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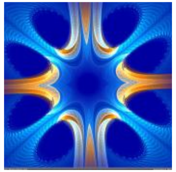
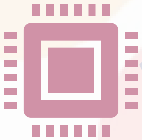
WHAT GEN AI?

Generative AI : Gen AI, Gen Models



generate new, original content that <u>resembles</u> human-created data.

utilize deep learning algorithms and neural networks



allows them to generate new content that is <u>coherent</u>, <u>contextually</u> relevant, and often <u>indistinguishable</u> from human-produced data.

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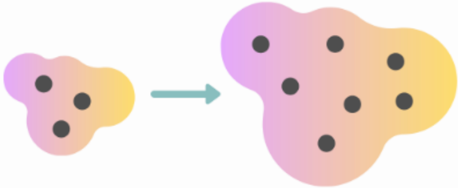
WHAT ARE GENERATIVE MODELS AND THEIR IMPORTANCE IN AI?

Generative models learn to generate new data similar to the training data

are essential in various fields such as **image** and **text generation**, **anomaly detection**, and **data augmentation**.



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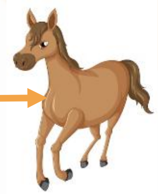
GENERATIVE - EXAMPLE (IMAGES)

- Generative modeling can be used to generate new images that resemble a specific category, even if they have never existed before.
- For example, in the case of a <u>dataset of horse images</u>

Training Data → Generative Model → Generated image →



Random noise → Generative Model

A sample data

One pixel of the sample with RGB value = [231, 134, 45]

- model learns the general rules that govern the appearance of a horse. (training)
- We pass some random data to the model (to predict)
- model can then generate new images of horses that look realistic based on the learned rules.

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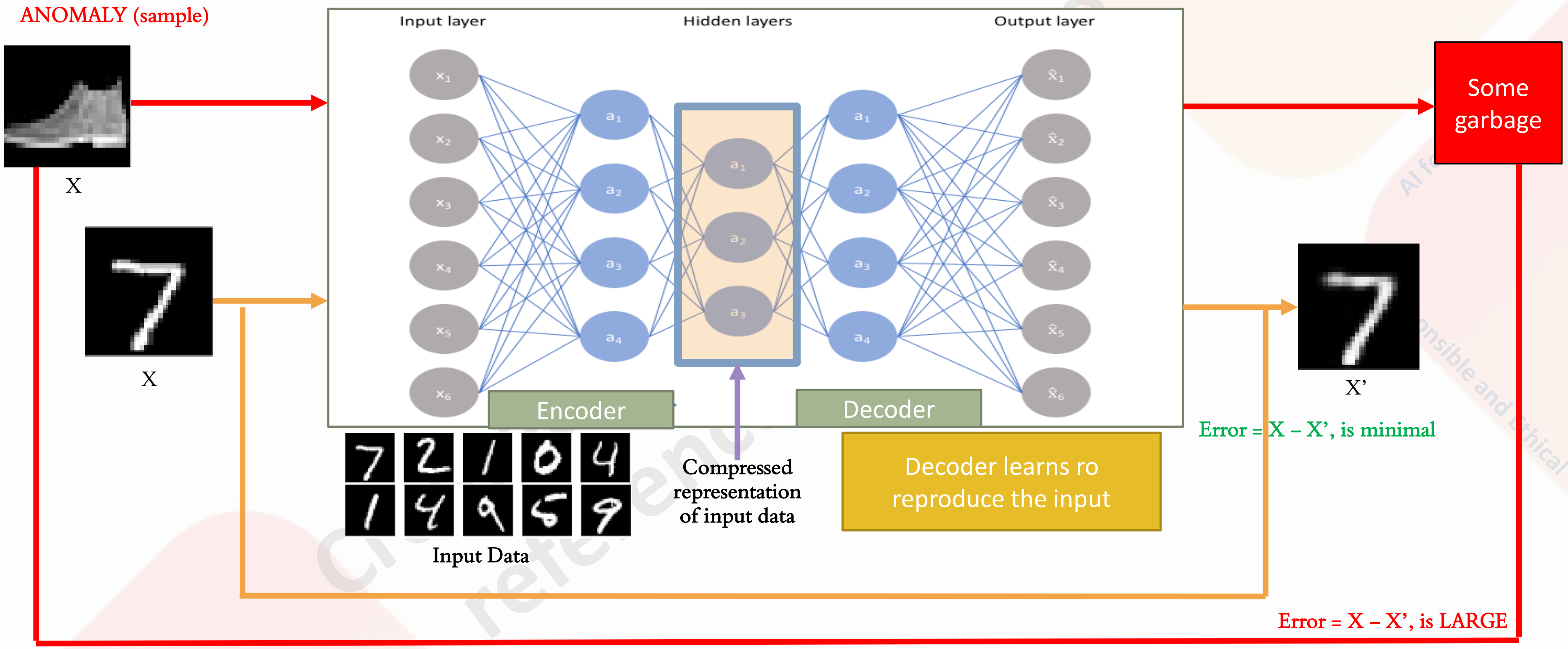
GENERATIVE - EXAMPLE (ANOMALY DETECTION)



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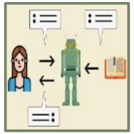
APPLICATIONS OF GENERATIVE MODELS IN AI

GAN-based image generation for creative design and artwork

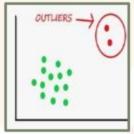
widely used to generate high-quality images, such as paintings, photographs, and faces.



Text generation
Recurrent Neural Networks (RNNs) and Transformer Networks are used to generate realistic and coherent text, such as news articles, product descriptions, and chatbot conversations.

Data synthesis for model training and testing

Variational Autoencoders (VAEs) and other generative models can be used to generate synthetic data for augmenting the training set



Novelty detection for fraud detection and cybersecurity

Generative models can be trained to learn the normal patterns of data and detect anomalous instances, such as fraudulent transactions and malicious activities.

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Question Generation

- If you have a dataset of statements or paragraphs, you can use an LLM to generate questions related to the content.
- This can be valuable for tasks like question-answering systems or reading comprehension, where having diverse questions is essential for training robust models.

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Data Masking and Filling

- LLMs can be used to mask certain parts of a text and predict what should go in those masked positions.
- This can be applied for data augmentation by creating partially masked sentences and then using the LLM to fill in the blanks.
- This is useful for tasks like text completion or language modeling.

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LANGUAGE COMPLETION

Given a partial sentence, the model can generate the missing words to complete it.

For example:

Prompt: "The sun was shining, the birds were singing, and the flowers were in full..."

Generated Text: "bloom. The vibrant colors painted the landscape, creating a picturesque scene of nature's beauty."

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Text Summarization:

Use an LLM to generate summaries of longer text documents.

These summaries can be considered as augmented data for tasks that require shorter text input, such as document classification or information retrieval.

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SUMMARIZATION

The model can generate concise summaries of longer texts.

Prompt: "Exercise offers a wide range of benefits for both physical and mental well-being. Improved Cardiovascular Health: Exercise strengthens the heart and improves circulation, leading to a reduced risk of heart disease, high blood pressure, and stroke. Regular physical activity helps in managing weight by burning calories, increasing metabolism, and building muscle mass.“

Generated Text: "Regular exercise provides numerous benefits for physical and mental well-being, including improved cardiovascular health, weight management through calorie burning and increased metabolism."

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Text Paraphrasing

LLMs are capable of paraphrasing sentences while preserving the original meaning.

This can be useful for generating variations of your text data.

For example, you can use an LLM to create paraphrased versions of your text for data augmentation in tasks like text summarization or sentiment

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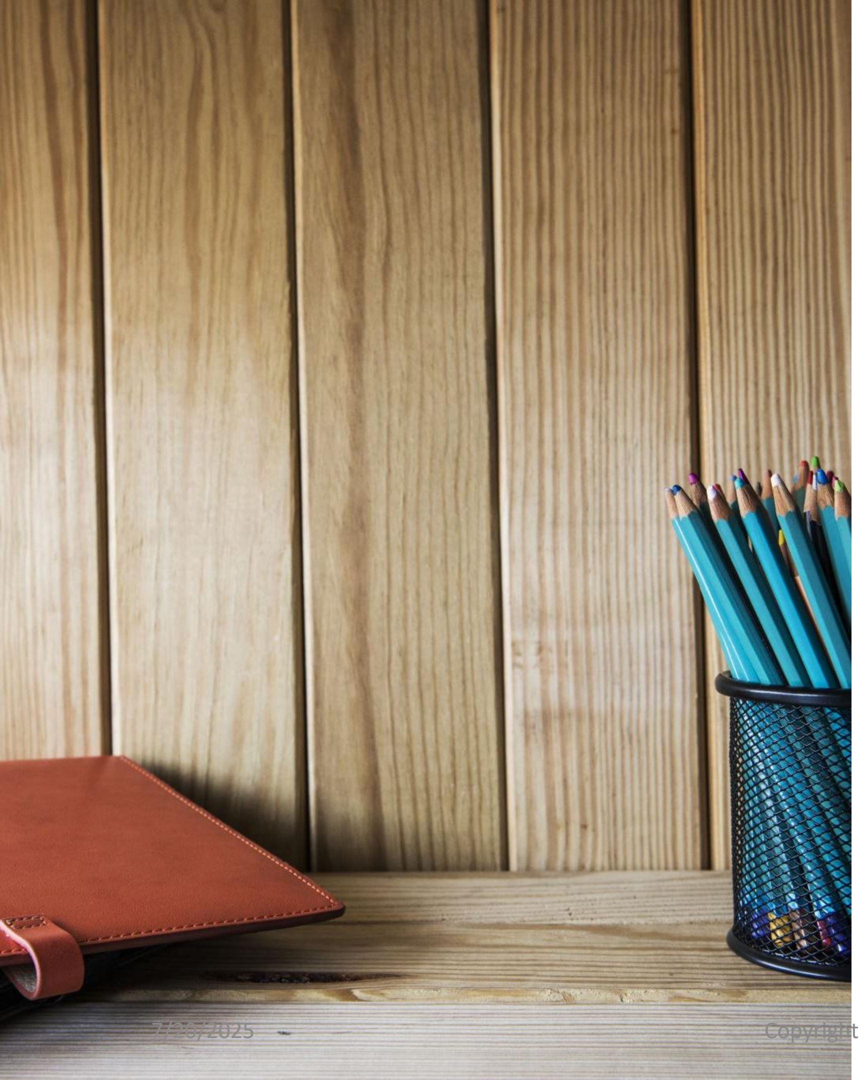

What is text paraphrasing

- Text paraphrasing is the process of rephrasing or restating a piece of text in your own words while retaining its original meaning.
- It involves expressing the content of a source text using different words and sentence structures while keeping the core ideas intact.

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Why use text paraphrasing?

Avoid	Improve	Enhance	Summarize	Maintain
Plagiarism: you can use information from a source while giving proper credit without directly copying the original text.	**Clarity**: Paraphrasing can make complex or convoluted sentences easier to understand by simplifying language and structure.	**Readability**: It allows you to adapt content to a specific audience or reading level, making it more accessible.	**Information**: Paraphrasing can be used to create concise summaries of longer texts, capturing the main points and key ideas.	**Original Meaning**: Effective paraphrasing should convey the same ideas and concepts as the original text while using different words.

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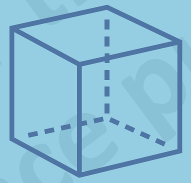
EXAMPLE (TEXT GENERATION)

Text generation models are computer programs that use natural language processing and machine learning to generate human-like text, with the goal of mimicking human language patterns and producing coherent text.

Inputs

Once upon a time,



Text Generation Model

Output

Once upon a time, we knew that our ancestors were on the verge of extinction. The great explorers and poets of the Old World, from Alexander the Great to Chaucer, are dead and gone. A good many of our ancient explorers and poets have

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TEXT GENERATION - APPROACHES

Rule-Based Generation

Requires
rules, templates, or patterns
Enables
Consistent, structured output
Applied in
generating greetings,
formulating responses, or
creating structured content.
Limitations
Lack of Flexibility
Difficulty Handling Ambiguity
No learning from Data
Maintenance

Markov Chain Text Generation

Requires
Based on probabilistic model
Enables
Variability in output
Applied in
Text Completion
Chatbots Agents
Text Augmentation
Limitations
Long-term Dependency
Repetitive/Predictable Output
Limited semantics
Sensitive to training data

Recurrent Neural Networks (RNN)

Requires
Sequence NN Modeling
Enables
capture sequential dependencies
Applied in
Text Completion
Chatbots Agents
Text Augmentation
Limitations
Long-term Dependency
Limited semantics
Repetition
Sensitive to training data
Computational Complexity

Transformers (e.g., GPT-3)

Requires
Transformer arch
Enables
Contextual Generation
Controllable Generation
Applied in
Text Completion
Chatbots Agents
Text Augmentation
Limitations
Computational Complexity
Training Data Requirements
Exposure Bias
OOV Words
Contextual Mistakes

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RULE-BASED GENERATION

Template-based:

Template: "Dear [Name], I hope this [occasion] brings you [adjective] and [adjective] moments. Wishing you [desire].“

Generated Text: "Dear Sarah, I hope this birthday brings you joy and memorable moments. Wishing you happiness and fulfillment."

Dear [Name], I hope this special day brings you memorable and joyful moments. Wishing you love, happiness, and all your heart's desires.

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CODE GENERATION

The model can generate code snippets based on specific programming languages.

For example:

Prompt: "Python code to calculate the factorial of a number:"

Generated Text:

python Copy code

```python
def factorial(n):
    if n == 0 or n == 1:
        return 1
    else:
        return n * factorial(n - 1)
```


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ML TO MODERN DAY GEN AI

1950s-1960s
Early AI Concepts, foundations of AI were laid down

1970s-1980s
Expert Systems: AI research focused on rule-based expert systems,

1990s
Neural Networks and Backpropagation: **Neural networks**, which are the basis for many generative models,

2000s
Deep Learning Resurgence, laid the groundwork for the development of generative models.

2010s
Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs)

2014
Text Generation: Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks gained popularity for text generation tasks.

2020s
Continued Advancements: The 2020s have seen ongoing advancements in generative AI. Models like GPT-3, with

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JOURNEY FROM TEXT TO GENERATIVE AI

2010

Near Perfect translation of Natural Language.

2014

Language models began to make sense of words.

2021

Foundation models introduced, used for fine-tuning.

2022

Arrival of true foundation model in GPT-3.5, and many more

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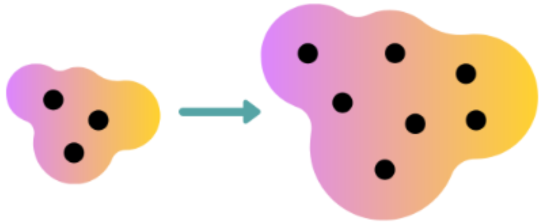

SUMMARIZE GENERATIVE MODELS

Generative models learn to generate new data similar to the training data

are essential in various fields such as **image** and **text generation**, **anomaly detection**, and **data augmentation**.

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TRANSFORMERS - SWING

- 2017 — **Transformer** Architecture:
- 2018 — Google **BERT** (Bidirectional Encoder Representations from Transformers)
- 2019 — **GPT-2** (Generative Pre-trained Transformer 2):
- 2020 — **T5** (Text-to-Text Transfer Transformer): Google AI introduced the T5 model
- 2020 — **BERT** Variants and Fine-tuning:
- 2020 — **GPT-3** (Generative Pre-trained Transformer 3):
- 2021 — **CLIP** (Contrastive Language–Image Pre-training):
- 2021 — **DALL-E** and Other Creative Applications: OpenAI introduced DALL-E,
- 2022 — **GPT-3.5** and Continued Progress:

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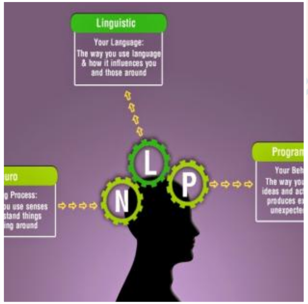
DETAILED LIST OF LLMS

Only a few ☺	
• BERT (Bidirectional Encoder Representations from Transformers) • GPT-3 (Generative Pre-trained Transformer 3) • RoBERTa (A Robustly Optimized BERT Pretraining Approach) • XLNet (eXtreme MultiLabel Network) • ALBERT (A Lite BERT) • DistilBERT (Distill BERT) • ELECTRA (Efficiently Learning an Encoder that Classifies Token Replacements Accurately) • T5 (Text-to-Text Transfer Transformer) • CTRL (Conditional Transformer Language Model) • OpenAI GPT (Generative Pre-trained Transformer)	• GPT-2 (Generative Pre-trained Transformer 2) • Transformer-XL • UniLM (Unified Language Model) • Longformer • SpanBERT • DeBERTa (Decoding-enhanced BERT with Disentangled Attention) • XLM-RoBERTa (Cross-lingual Language Model - RoBERTa) • MT-DNN (Multi-Task Deep Neural Networks) • CamemBERT (Contextualized Language Model for French) • SciBERT (Scientific Text BERT) 

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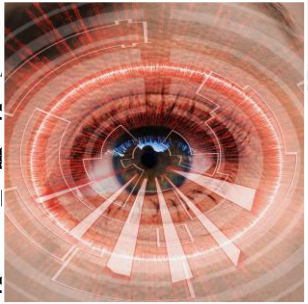
DETAILED LIST LLMS

Only a few ☺

- DALL-E: Text-to-image generator.
- CLIP: Vision-language multimodal understanding model.
- BigGAN: High-resolution GAN image synthesis.
- StyleGAN2: Fine-grained style-controlled image generation.
- VQ-VAE-2: Disentangled image representation learning.
- ProGAN: Progressive high-quality GAN images.
- pix2pixHD: High-resolution image-to-image translation.
- GPT-2: Text-based image generation extension.
- AttnGAN: Attention-based GAN image synthesis.
- CycleGAN: Unpaired image-to-image translation.
- UNIT: Unsupervised image translation model.
- SPADE: Semantic image synthesis with normalization.
- pix2pix: Conditional GAN image translation.
- SAGAN: GANs with self-attention mechanism.
- DeepArt.io: Neural style transfer service.

- DeepDream: Image hallucination and stylization.
- FastStyleTransfer: Real-time artistic style transfer.
- Neural Style Transfer: Apply artistic styles to images.
- Neural Doodle: Combine images creatively.
- PaintsChainer: Automatic colorization of sketches.
- Artbreeder: Blend and morph images creatively.
- NeuralTalk2: Image captioning with attention mechanism.
- Selfie2Anime: Turn selfies into anime-style avatars.
- ESRGAN (Enhanced Super-Resolution GAN): Super-resolution for images.
- SRGAN (Super-Resolution Generative Network): Super-resolution with GAN
- StarGAN: Multi-domain image-to-ima
- BiGAN (Bidirectional Generative Adve Bidirectional image-to-image translati
- StackGAN: Text-to-image synthesis in
- SeqGAN: Generate images from text descriptions.



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- Discriminative Vs Generative Models

DEFINE DISCRIMINATIVE ML

is a type of statistical model that focuses on modeling the relationship between the input data (features) and the output variable (target or label).

The primary objective of a discriminative model is to learn the conditional probability distribution of the output variable given the input data.

In other words, it aims to find the decision boundary or decision function that separates different classes or categories in the data.

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Problem statement – Customer Segmentation

Definition

- process of dividing a customer base into distinct groups or segments based on shared characteristics, behaviors, or demographics.

Objective

- helps businesses tailor their marketing, product offerings, and services to better meet the specific requirements of each segment.

Methods

- Demographic Segmentation
- Behavioral Segmentation
- Geographic Segmentation

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Data setup

Introduction:

- four customer classes based on income and spend predictors.

Predictor Columns:

- Column 1: Income
- Column 2: Spend

Income and Spend Ranges:

- Income: 1000 to 100000
- Spend: 1000 to 100000

Customer Classes:

- ES (Economically Stressed): Low income, Low spend
- SC (Savings Customer): High income, Low spend
- RC (Risk Customer): Low income, High spend
- PC (Prudent Customer): High income, High spend

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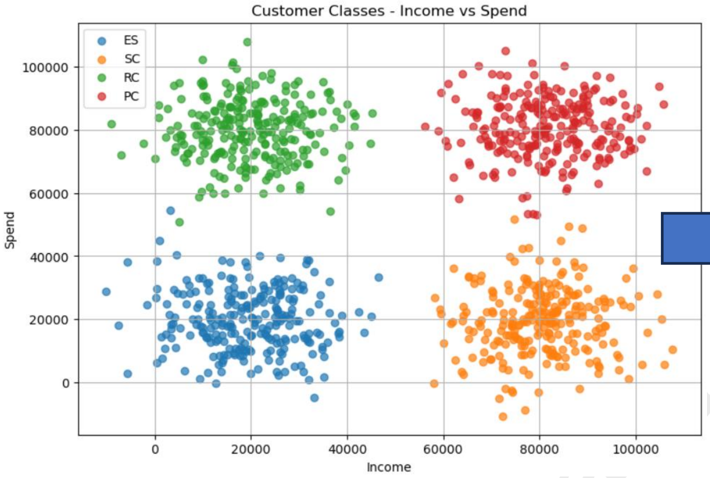
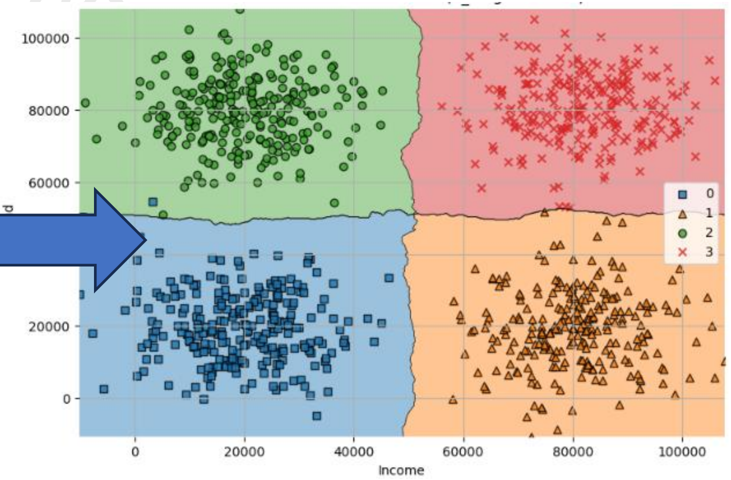
Machine learning – Input and output





Input Data

Income	Spend	Customer Type

Learning from Data

Map {Income, Spend} to {Customer Type}

Output (Model)

Classified data using class boundaries

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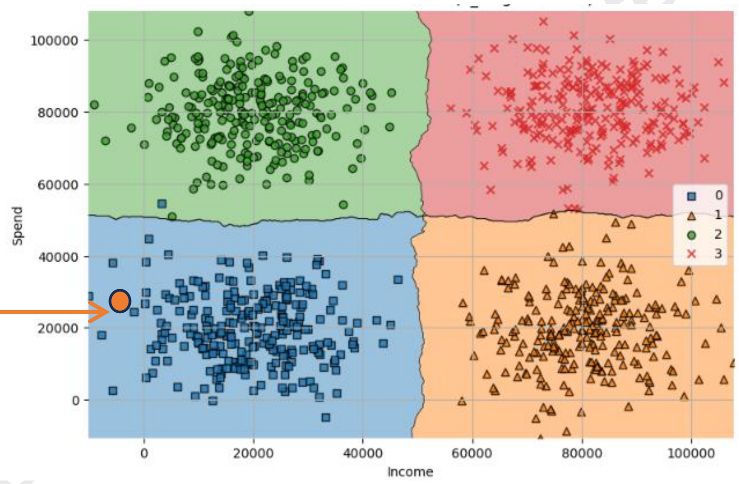
How the decision boundaries help?

Predict customer type of a New customer

Income	Spend
32805.917934	9034.764100



Output (Model)

Classified data using class boundaries

Prediction (customer type)

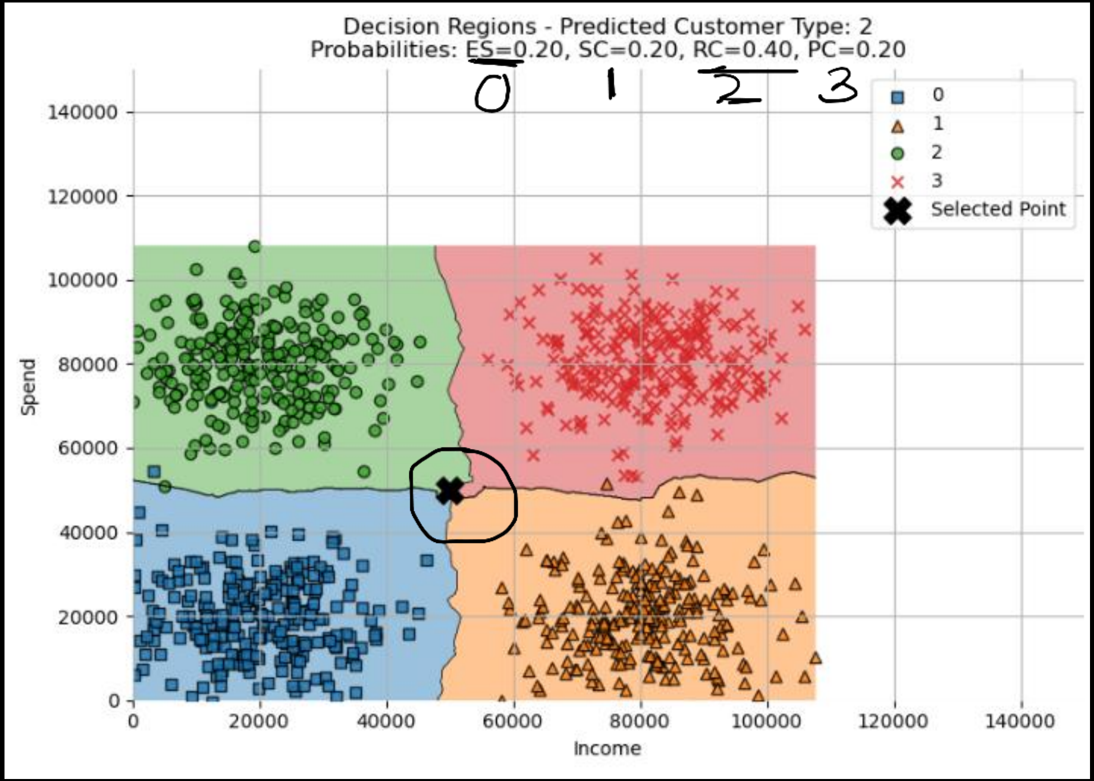
Class	target
ES	0

Predicted Class: [0]

Prediction Probabilities: [[0.6, 0.4, 0., 0.]]

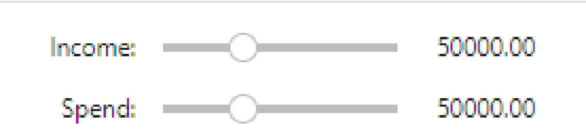
i.e.
New customer is closer/ more similar to customers of ES type than other types

Typical prediction



- Valid input data
  - Income : 0 to 100,000
  - Spend : 0 to 100,000

- New customer data



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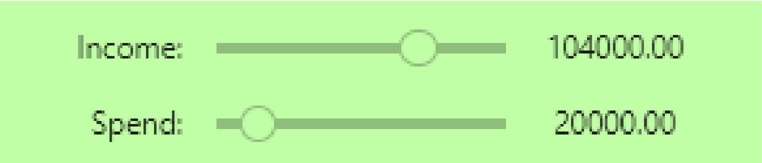
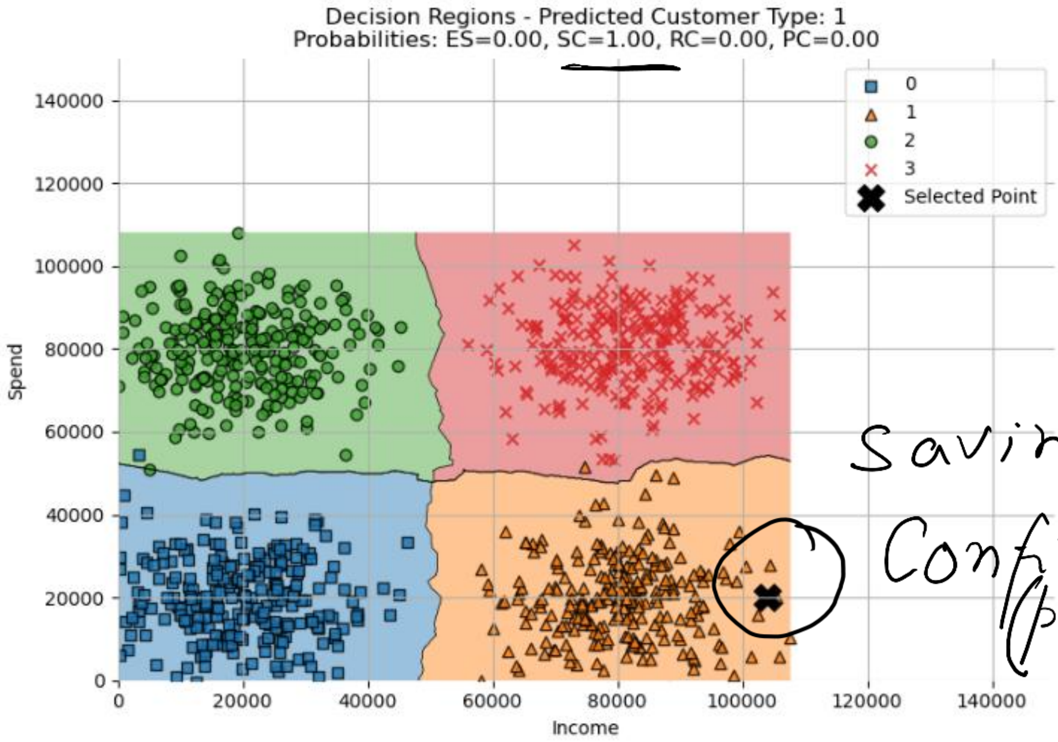
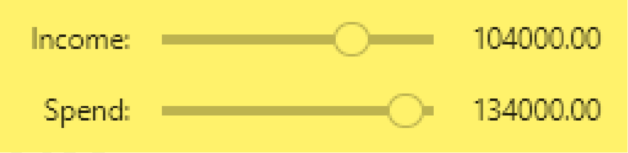
Another customer

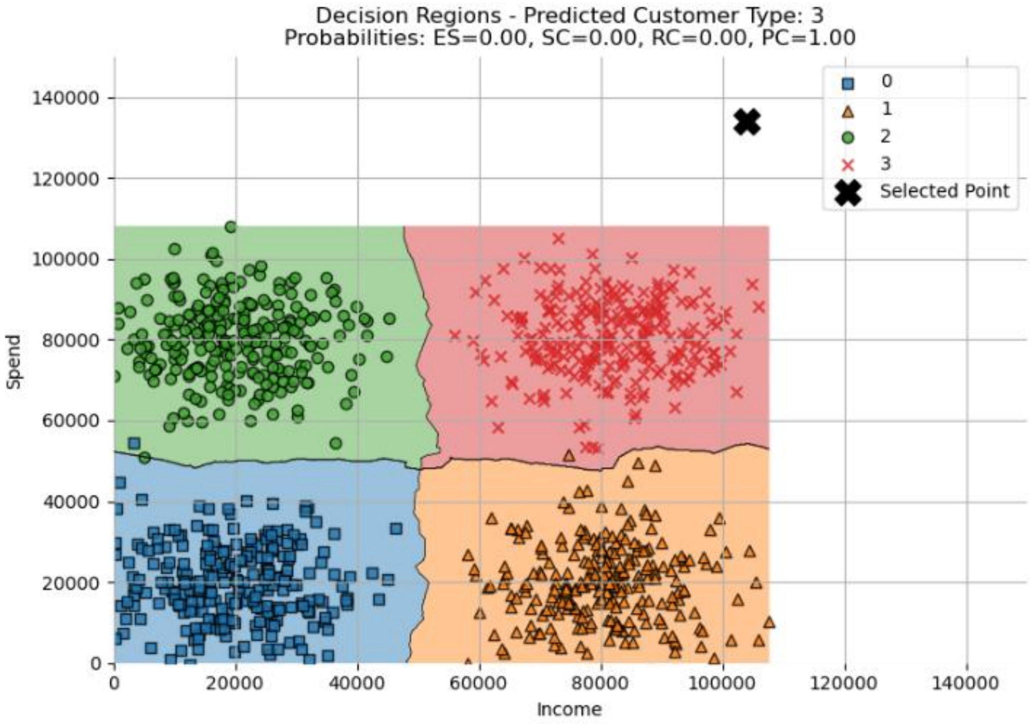




Savings Type
Confident
(pred prob)

Odd customer





Validation of Input data for prediction is required

(use of inference pipeline)

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what if the prediction probabilities are same?

Model Uncertainty

- close probabilities suggest that the input might lie near the decision boundary between classes
- making it challenging for the model to confidently assign it to one specific class.

Feature Ambiguity

- input features may not provide clear distinctions between the classes
- Could mean noisy data, possibly

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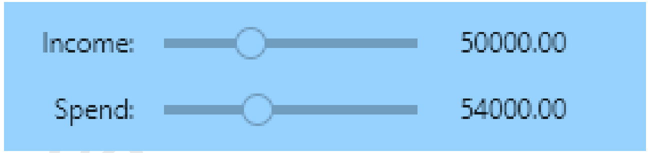
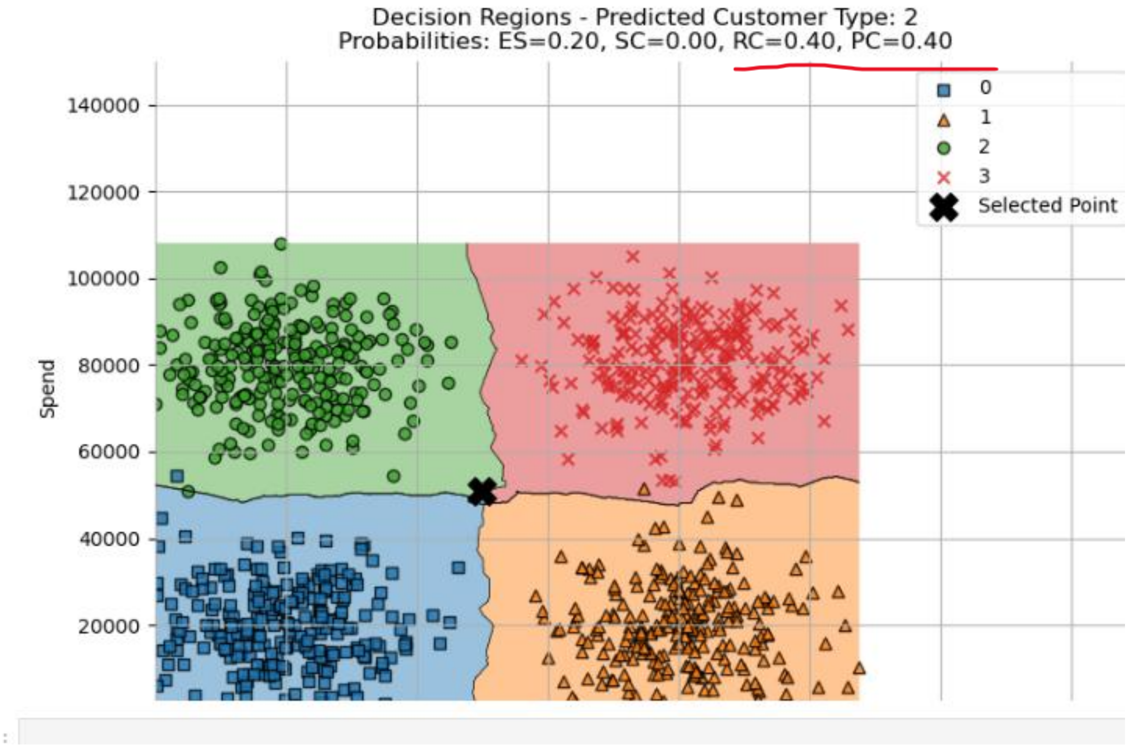
Example





ML algorithms use various different heuristics to break the tie

(some models do arbitrarily too)

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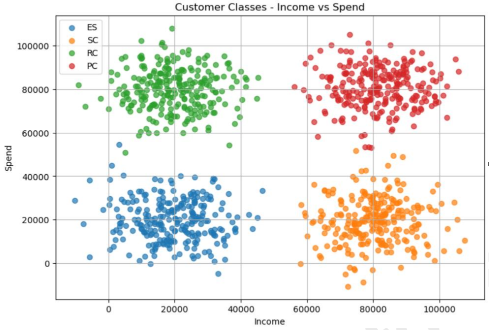
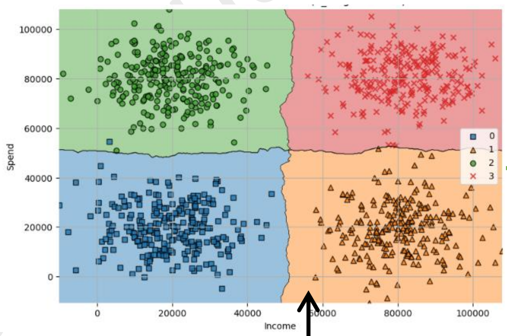
But how good are the class boundaries? How to check?

Training Data :



ML Model:



Test Data :

	Income	Spend	Class	target
251	65284.796931	16704.322641	SC	1
536	7619.595991	85279.986831	RC	2
284	90165.273152	24076.408727	SC	1
111	8401.540579	28816.735024	ES	0
700	25389.476934	82954.129496	RC	2
502	14324.214015	85275.182847	RC	2

Actual label

Income	Spend
65284.796931	16704.322641
7619.595991	85279.986831
90165.273152	24076.408727
8401.540579	28816.735024
25389.476934	82954.129496
14324.214015	85275.182847

Class	target
ES	0
ES	0
ES	0
SC	1
RC	2
PC	3

Predicted label

Compare

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Actual labels vs predicted labels

Test Data :
100 samples

	Income	Spend	Class	target
251	65284.796931	16704.322641	SC	1
536	7619.595991	85279.986831	RC	2
284	90165.273152	24076.408727	SC	1
111	8401.540579	28816.735024	ES	0
700	25389.476934	82954.129496	RC	2
502	14324.214015	85275.182847	RC	2

Actual label

Compare

Class	target
ES	0
ES	0
ES	0
SC	1
RC	2
PC	3

Predicted label

Number of predictions = 100

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}}$$

$$\text{Accuracy} = \frac{90}{100} = 0.9$$

So, the accuracy of the model in this scenario is 0.9 or 90%.

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